

Converting an Iterative Algorithm into a Recursive Algorithm

Likewise, converting an iterative algorithm into a recursive algorithm is always possible. An iterative algorithm is simply code that uses a loop. The code that is repeatedly executed (the loop's body) can be placed in a recursive function's body. And just as the code in the loop's body is executed repeatedly, we need to repeatedly call the function to execute its code. We can do this by calling the function from the function itself, creating a recursive function.

The Python code in *hello.py* demonstrates printing `Hello, world!` five times by using a loop and then also using a recursive function:

Python

```
print('Code in a loop:')
i = 0
while i < 5:
    print(i, 'Hello, world!')
    i = i + 1

print('Code in a function:')
def hello(i=0):
    print(i, 'Hello, world!')
    i = i + 1
    if i < 5:
        hello(i) # RECURSIVE CASE
    else:
        return # BASE CASE
hello()
```

The equivalent JavaScript code is in *hello.html*:

JavaScript

```
<script type="text/javascript">
document.write("Code in a loop:<br />");
let i = 0;
while (i < 5) {
```